

LoFi Granulator – User Guide

The LoFi Granulator is an instrument that takes an audio sample from an SD card and cuts it into small fragments for playback. These fragments can be played in order or out of order, forwards or backwards, tuned higher or lower, and muted or unmuted.

Getting started:

- Ensure that the SD card with WAV files is inserted.
- Power the Granulator via USB or batteries and turn on.
- It may take a few seconds for the sounds to load from the SD card.
- Press the **Green button** to start playback. Press again to stop.
- Playback loops around the sample such that sound always plays.

Changing sounds:

The Granulator selects the first (alphabetical) WAV file on the SD card at startup. To select the next file, **hold the Blue button and press the Green button**. Repeat this to cycle through successive files; the selection will loop back to the start at the end of the list.

Randomising grain parameters:

An easy way to explore the possibilities of the LoFi Granulator is to **click the Blue button** to randomise the parameters.

Changing the grain parameters:

There are 4 dials on the LoFi Granulator. These adjust 8 grain parameters, four directly and another 4 when the Blue button is held down.

Dials are numbered 1-4 from the bottom left corner, counting clockwise.

- Dial 1 – Grain Size / Grain Start Position diversity
- Dial 2 – Grain Pitch / Grain Pitch diversity
- Dial 3 – Playback Position within the sample / Playback direction and speed (0 in the middle)
- Dial 4 – Grain Density / Volume

Adding sounds to the SD card:

The LoFi Granulator will play any valid WAV file up to its memory limit. This is approximately 10 seconds of mono audio at 44.1k, 16bit sample rate. If longer files are

used, the Granulator will load as much of it as possible from the start and truncate the rest.

Many audio sample rates and bit depths are supported, but 16bit, 44.1k is optimal.

Load files onto the SD card from your computer. The SD card is formatted as FAT32, so should be readable on a PC, Mac or Linux computer.

Experiment with different sound sources to explore the variety of sonic possibilities of the Granulator.

Effects:

The LoFi Granulator includes some effects, delay and reverberation.

- Subtle reverberation is always on.
- Delay amount increases as the density decreases.

Problem solving:

If there is no sound, then;

- Power down and restart the Granulator.
- Check the SD card is inserted and has WAV files on it.
- Ensure the volume is turned up.
- See if any internal wire connections have dislodged.

Updating:

The documents about the LoFi Granulator are available on GitHub, including the assembly instructions and latest software update:

<https://github.com/algomusic/lofi-granulator>